

NPDES PERMIT REVIEW SUMMARY

Facility:	Cadillac WWTP
Permit Number:	MI0020257
Reviewer:	Glen Schmitt, Lakes Michigan and Superior Permits Unit, Permits Section, WRD, MDEQ
Date:	1/30/2018 (updated on 4/11/2019)

Outf.	Authorized Flow		Disch. Type *	Receiving Water	Lake?	FLOWS (cfs)			pH	Hardness
	mgd	cfs				95% Exc.	H. Mean	90Q10		
001	3.2	4.95		Clam River	☐	2.9	14.0	3.6	7	75

* P = Process; N = Noncontact Cooling; C = Contact Cooling; S = Sanitary; ST = Storm Water; GW = Groundwater Purge

SOURCES REVIEWED

X	Permit Application	Type of issuance: REISSUANCE
X	Facility/VGW File	Low Flow File No. 8970
X	MiWaters Documents & Files	Hardness value collected from Clam River during 10/14/2014 Point Source Monitoring Survey
X	Current Permit	Parameters Limited:

Parameter	Mo Avg.	Load	D Max

Special Conditions:

WTA

WTA/NMS GUIDANCE

X	Biosurvey Reports	Report #:
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X	WET Tests	Date:
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X	MiSWIMS/STORET
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REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 1 of 2)

*concentration values in ug/L except as noted

Facility:	Cadillac WWTP	Hardness:	75 mg/L CaCO3
Outfall:	001	pH:	7
Date:	1/30/2018 (updated on 4/11/2019)	Drinking Water:	n

Parameter	CAS #	Water Quality Values										Diss. Met. Translator	Background Conc.
		FCV	vd	HNV	vd	HCV	vd	WV	vd	FAV	vd		
Cyanide, free	57125	5.2	1199707	48000	1199707	NA	0	NA	0	44	1199707	1	0
Copper	7440508	7.00393125	1199707	38000	1200512	NA	0	NA	0	20.49667108	1199707	1.5	4
Mercury @	7439976	0.77	1199707	0.0018	1199707	NA	0	0.0013	1199707	2.8	1199707	1	**
Barium	7440393	322.4532343	2200905	160000	1199705	NA	0	NA	0	1840.068569	2200905	1	0
Boron	7440428	7200	1201511	330000	1201511	NA	0	NA	0	69000	1201511	1	0
Cadmium	7440439	1.809112735	1199707	130	1199706	NA	0	NA	0	6.243962757	1199707	2.1	0
Chromium	7440473	58.55695529	1199707	9400	1199706	NA	0	NA	0	900.3259642	1199707	1.5	1.6
Lead	7439921	16.50732835	2201011	190	1200709	NA	0	NA	0	316.5089773	1201011	4.5	2.5
Molybdenum	7439987	3200	2200604	10000	1200605	NA	0	NA	0	58000	2200604	1	0
Nickel	7440020	40.77179058	1199707	210000	1199706	NA	0	NA	0	734.1696421	1199707	1.1	3
Selenium	7782492	5	1199707	2700	1199704	NA	0	NA	0	120	1199808	1	0
Silver	7440224	0.06	1199710	11000	1199705	NA	0	NA	0	1.1	1199710	20	0
Zinc	7440666	92.58329763	1199707	16000	1200510	NA	0	NA	0	183.6642294	1199707	2.1	13
Bis(2-ethylhexyl)phthalate #	117817	ID*	199809	160	1199711	18	1201402	NA	0	285	2199809	1	0
#N/A	1	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
#N/A	0	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A

= Carcinogen

** = Background exceeds standards.

REASONABLE POTENTIAL: CHEMICAL SPECIFIC (p. 2 of 2)

**concentration values in ug/L except as noted; loads in lb/d*

Facility:	Cadillac WWTP	Disch. Rate:	4.95 cfs	95% Ex. Flow:	2.9 cfs	Conc.	Load (lbs/day)
Outfall:	001			H. Mean Flow:	14 cfs	Hg Loading Calculated based on LCA (ng/L) of :	0
Date:	1/30/2018 (updated on 4/11/2019)			90Q10 Flow:	3.6 cfs	TP Loading based on concentration (mg/L) of :	0

Parameter	Monthly Average PEL								Daily Max PEL		PEQ		DECISION	
	FCV	load	HNV	load	HCV	load	WV	load	conc	load	Avg	Max	Avg	Max
Cyanide, free	5.96106875	0.159089	81915	2186.1475	#VALUE!	#VALUE!	#VALUE!	#VALUE!	44	1.174272	5.5	5.5	No Reasonable Potential. Remove Limit and monitoring requirements.	
Copper	11.45809587	0.3057937	64846.54875	1730.6247	#VALUE!	#VALUE!	#VALUE!	#VALUE!	30.745007	0.8205227	180.6	180.6	Reasonable potential. Recommend a Mon. Avg. and Daily Max. WQBEL of 11 ug/l (0.3 lbs/day) and 31 ug/l (0.82 lb/day), weekly monitoring.	
Mercury @	0.77	0.0205498	0.0018	4.804E-05	NA	#VALUE!	0.0013	3.46944E-05	2.8	0.0747264	0.00154	0.00154	Reasonable Potential. Recommend retain current LCA and monitoring requirements and PMP.	
Barium	369.6472882	9.8651468	273050	7287.1584	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1840.0686	49.10775	66.7	66.7	No Reasonable Potential. No recommendations.	
Boron	8253.7875	220.27708	563165.625	15029.764	#VALUE!	#VALUE!	#VALUE!	#VALUE!	69000	1841.472	1378	1378	No Reasonable Potential. No recommendations.	
Cadmium	4.355176023	0.1162309	221.853125	5.9208162	#VALUE!	#VALUE!	#VALUE!	#VALUE!	13.112322	0.3499416	1.104	1.104	No Reasonable Potential. No recommendations.	
Chromium	100.456797	2.680991	16040.557	428.09039	#VALUE!	#VALUE!	#VALUE!	#VALUE!	1350.4889	36.041849	3.352	3.352	No Reasonable Potential. No recommendations.	
Lead	84.78908932	2.2628512	322.4804688	8.6063588	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples 1424.2904	38.011462	2.1199587	4.5378898	No Reasonable Potential. No recommendations.	
Molybdenum	3668.35	97.900925	17065.625	455.4474	#VALUE!	#VALUE!	#VALUE!	#VALUE!	58000	1547.904	4.98	4.98	No Reasonable Potential. No recommendations.	
Nickel	50.97395868	1.360393	358376.0053	9564.3388	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples 807.58661	21.552871	7.7114224	12.764735	No Reasonable Potential. Recommend Quarterly Monitoring due to elevated concentrations and concerns for Total Nickel.	
Selenium	5.731796875	0.1529702	4607.71875	122.9708	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples 120	3.20256	10.502357	35.819419	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 5.7 ug/l (0.15 lbs/day), monthly monitoring.	
Silver	1.37563125	0.0367128	18772.1875	500.99214	#VALUE!	#VALUE!	#VALUE!	#VALUE!	22	0.587136	2.47	2.47	No Reasonable Potential. No recommendations.	
Zinc	220.9781637	5.8974652	27295.81469	728.4707	#VALUE!	#VALUE!	#VALUE!	#VALUE!	PEQ >= +10 Samples 385.69488	10.293425	0.7193668	1.5553517	No Reasonable Potential. No recommendations.	
Bis(2-ethylhexyl)phthalate #	#VALUE!	#VALUE!	273.05	7.2871584	30.718125	0.8198053	#VALUE!	#VALUE!	285	7.60608	49.4	49.4	Reasonable Potential. Recommend a Mon. Avg. WQBEL of 31 ug/l (0.82 lbs/day), weekly monitoring.	
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		
#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A	0	0		

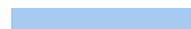
* SILVER - Appropriate Dissolved Metal Translator must be entered in PEL 1

PEQ -- Effluent Data Summary
(all values in ug/L)

DATA ENTRY

[illegible]

PEQ -- Effluent Data Summary
(all values in ug/L)



Parameter:	Nickel	Source	Selenium	Source	Silver	Source	Zinc	Source	Bis(2-ethylhexyl)phthalate #	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A	Source
No. Nondetect	104	Ipp Data/App.	55	DMRs	5	App./MiWaters	0	DMRs/App.	3	App.	0		0		0		0	
Data	5.1	5/2/2016	1.9	5/2/2016	0.2	Jan-15	51.2	May-15	19	5/2/2016								
	5.5	8/1/2016	6.6	8/1/2016	0.55	Feb-15	55	Jun-15										
	5	2/2/2017	4	10/6/2016	1.55	Mar-15	40	5/2/2016										
	11.7	5/21/2014			1.58	Apr-15	35	8/1/2016										
	8.11	5/27/2014			0.77	Jun-15	29	10/6/2016										
	50	2/9/2016			0.66	Jul-15	22	2/2/2017										
	78	5/3/2016	QL <1.0 to <2.0 ug/l		0.34	Aug-15	20	5/9/2018		Not Detected								
	92	5/13/2016			1.18	Sep-15				8/1/2016 (<10 ug/l)								
	56	11/1/2017			0.26	Jun-16				10/6/2016 (<10 ug/l)								
	58	3/9/2018	Not Detected		0.02	Jul-16				2/2/2017 (<5.0 ug/l)								
	68	5/1/2018		6/5/2013	0.37	Oct-16		There was										
	65	5/11/2018		6/8/2013	1.14	Dec-16		several zinc										
	51	9/14/2018		6/11/2013				data points										
	4.5	5/9/2018		6/19/2013				with a result										
				6/28/2013				of "0". These										
				7/2/2013				data were not										
				7/8/2013				used since the										
				7/17/2013				QL could										
				7/20/2013				not be verified.										
	Rest of dates used a			7/25/2013														
	QL <50 ug/l			7/30/2013	Not Detected													
				8/7/2013		5/1/2016												
	Not Detected			8/10/2013		8/1/2016												
	10/6/2016 (<5.0 ug/l)			8/14/2013		10/6/2016												
				8/22/2013		2/2/2017												
				8/27/2013		5/9/2018												
				9/4/2013														
				9/7/2013	QL <0.2 to <0.5 ug/l													
				9/11/2013														
				9/20/2013														
				9/24/2013														
				10/1/2013	There was													
				10/4/2013	several silver													
				10/8/2013	data points													
				10/16/2013	with a result													
				10/24/2013	of "0". These													
				10/28/2013	data were not													
				11/4/2013	used since the													
				11/8/2013	QL could													
				11/12/2013	not be verified.													
				11/19/2013														
				11/26/2013														
				12/3/2013														
				12/10/2013														
				12/13/2013														
				12/18/2013														
				12/26/2013														
				1/2/2014														
				1/5/2014														
				1/6/2015														
				1/7/2014														
				1/9/2015														
				1/22/2014														
				1/30/2014														
				2/6/2014														
				2/11/2014														
				2/14/2014														
				2/19/2014														
				2/24/2014														
				3/4/2014														
				3/7/2014														
				3/11/2014														
				3/19/2014														

PEQ -- Effluent Data Summary
(all values in ug/L)

Parameter:	Nickel	Source	Selenium	Source	Silver	Source	Zinc	Source	Bis(2-ethylhexyl)phthalate #	Source	#N/A	Source	#N/A	Source	#N/A	Source	#N/A	Source
		11/6/2015		3/25/2014														
		12/1/2015		2/2/2017														
		12/11/2015		5/9/2018														
		1/8/2016																
		1/15/2016																
		2/12/2016																
		3/1/2016																
		4/5/2016																
		4/8/2016																
		6/1/2016																
		6/10/2016																
		7/6/2016																
		7/15/2016																
		8/2/2016																
		8/12/2016																
		9/6/2016																
		9/9/2016																
		10/4/2016																
		10/7/2016																
		11/1/2016																
		11/18/2016																
		12/7/2016																
		12/9/2016																
		1/3/2017																
		1/13/2017																
		1/31/2017																
		2/10/2017																
		2/28/2017																
		3/10/2017																
		4/4/2017																
		4/7/2017																
		4/30/2017																
		5/12/2017																
		6/7/2017																
		6/9/2017																
		7/4/2017																
		7/14/2017																
		8/1/2017																
		8/11/2017																
		9/5/2017																
		9/15/2017																
		10/3/2017																
		10/13/2017																
		11/17/2017																
		12/5/2017																
		12/15/2017																
		1/3/2018																
		1/19/2018																
		2/6/2018																
		2/16/2018																
		3/5/2018																
		4/3/2018																
		4/13/2018																
		6/5/2018																
		6/15/2018																
		7/2/2018																
		7/13/2018																
		7/31/2018																
		8/17/2018																
		9/4/2018																
		11/5/2018																
		11/9/2018																
		12/4/2018																
		12/14/2018																

PROJECTED EFFLUENT QUALITY ANALYSIS (<10 Detectable):

* all values in ug/l

	Cyanide, free	Copper	Mercury @	Barium	Boron	Cadmium	Chromium	Lead	Molybdenum	Nickel	Selenium	Silver	Zinc	Bis(2-ethylhexyl)phthalate #	#N/A	#N/A	#N/A	#N/A	#N/A	#N/A
Sample Max.	5	86	0.0011	29	530	0.69	8.47	2.65	8.11	92	6.6	1.58	55	19	0	0	0	0	0	0
Total Samples	48	6	17	5	4	12	12	11	9	113	58	14	7	4	0	0	0	0	0	0
Multiplier	1.1	2.1	1.4	2.3	2.6	1.6	1.6	1.7	1.8	0.9	1	1.5	2	2.6	0	0	0	0	0	0
PEQ:	5.5	180.6	0.00154	66.7	1378	1.104	13.552	4.505	14.598	82.8	6.6	2.37	110	49.4	0	0	0	0	0	0

PROJECTED EFFLUENT QUALITY ANALYSIS (>=10 Detectable):

PARAMETER:	Cyanide, free		Copper		Mercury @		Barium		Boron		Cadmium	
TOTAL NO. VALUES:	48		6		17		5		4		12	
DETECTED:	2		6		4		5		4		7	
NON-DETECTED:	46		0		13		0		0		5	
d (% data<detect)	0.958333333		0		0.764706		0		0		0.4166667	
m (mean of data >detect)	3.5		47.61667		0.000831		22.8		425		0.3171429	
std dev	2.121320344		33.80582		0.000312		3.701351		96.781541		0.17385	
n	1 30		1 30		1 4		1 30		1 30		1 30	
d^n	0.958333333	0.278932	0	0	0.764706	0.341962	0	0	0	0	0.4166667	3.923E-12
p	-0.2	0.930658	0.95	0.95	0.7875	0.924017	0.95	0.95	0.95	0.95	0.9142857	0.95
std normal dist fn (Z)	#NUM!	2.310286	2.447747	2.447747	1.760008	2.270348	2.447747	2.447747	2.4477468	2.4477468	2.2166352	2.4477468
Z_p	#NUM!	1.481002	1.645211	1.645211	0.797575	1.432884	1.645211	1.645211	1.6452114	1.6452114	1.3678576	1.6452114
1+(s/m)^2	1.367346939	1.367347	1.50404	1.50404	1.140728	1.140728	1.026354	1.026354	1.051857	1.051857	1.3004964	1.3004964
(sigma_d)^2	0.312872321	0.312872	0.408155	0.408155	0.131666	0.131666	0.026013	0.026013	0.0505572	0.0505572	0.2627461	0.2627461
mu_d	1.096326808	1.096327	3.659105	3.659105	-7.158413	-7.158413	3.113754	3.113754	6.0268106	6.0268106	-1.279776	-1.279776
(sigma_dn)^2	0.312872321	0.395949	0.408155	0.016662	0.131666	0.255484	0.026013	0.000878	0.0505572	0.0017271	0.2627461	0.0401633
mu_dn	1.096326808	-1.796244	3.659105	3.854852	-7.158413	-8.248748	3.113754	3.126321	6.0268106	6.0512256	-1.279776	-1.707481
P_95 exponent	#NUM!	-0.864331	4.710182	4.067217	-6.869006	-7.524491	3.379103	3.175073	6.396735	6.1195975	-0.578629	-1.377768
P_95 (PEQ)	#NUM!	0.421333	111.0723	58.3942	0.00104	0.00054	29.34443	23.92857	599.88324	454.68163	0.5606665	0.2521407
	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

NOTES: For purposes of this summary, ^ represents an exponent or superscript while _ represents a subscript.
Variables are defined in Part 12, Rule 1225 (3) (a).

Chromium		Lead		Molybdenum		Nickel		Selenium		Silver		Zinc	
23		11		24		118		58		17		7	
20		6		24		14		3		12		7	
3		5		0		104		55		5		0	
0.1304348		0.4545455		0		0.88135593		0.94827586		0.29411765		0	
1.8285		2.1183333		6.70708333		39.8507143		4.16666667		0.71833333		36.0285714	
1.6155699		0.2943071		3.20660254		31.7066604		2.35442845		0.53004002		13.6016806	
1	30	1	30	1	30	1	30	1	30	1	30	1	30
0.1304348	2.896E-27	0.4545455	5.337E-11	0	0	0.88135593	0.02262247	0.94827586	0.203254832	0.29411765	1.13667E-16	0	0
0.9425	0.95	0.9083333	0.95	0.95	0.95	0.57857143	0.9488427	0.03333333	0.937244677	0.92916667	0.95	0.95	0.95
2.3899667	2.4477468	2.1861365	2.4477468	2.44774683	2.44774683	1.314614	2.43838061	0.26039029	2.353088134	2.30105436	2.447746831	2.44774683	2.44774683
1.5764439	1.6452114	1.3307693	1.6452114	1.64521144	1.64521144	0.19786871	1.63408839	-1.70598055	1.532361055	1.46989691	1.64521144	1.64521144	1.64521144
1.7806594	1.7806594	1.0193025	1.0193025	1.22857191	1.22857191	1.63303654	1.63303654	1.319296	1.319296	1.54445913	1.544459132	1.142525	1.142525
0.5769837	0.5769837	0.0191185	0.0191185	0.20585245	0.20585245	0.49044119	0.49044119	0.27709826	0.277098261	0.43467377	0.434673772	0.13324073	0.13324073
0.3150041	0.3150041	0.7410703	0.7410703	1.80023796	1.80023796	3.43991973	3.43991973	1.28856723	1.288567225	-0.54815845	-0.548158451	3.51769191	3.51769191
0.5769837	0.0343292	0.0191185	0.028546	0.20585245	0.00759019	0.49044119	0.33162083	0.27709826	0.369900159	0.43467377	0.038835503	0.13324073	0.00473958
0.3150041	0.4465694	0.7410703	0.1302208	1.80023796	1.89936909	3.43991973	1.4105849	1.28856723	-1.492444056	-0.54815845	-0.698546011	3.51769191	3.58194248
1.5124621	0.7513966	0.9250756	0.4081887	2.54668627	2.04270266	3.57849017	2.35159968	0.39053696	-0.560470983	0.42094155	-0.374328707	4.11822953	3.69520646
4.5378898	2.1199587	2.5220588	1.5040909	12.7647347	7.71142238	35.8194187	10.5023567	1.47777408	0.570940098	1.52339524	0.687750806	61.45035	40.2538827
Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average	Max	Average

